

As per Feedback session instructions, below tasks were performed for selected countries which are Canada, Denmark, United States and Wales.

Task D.1:

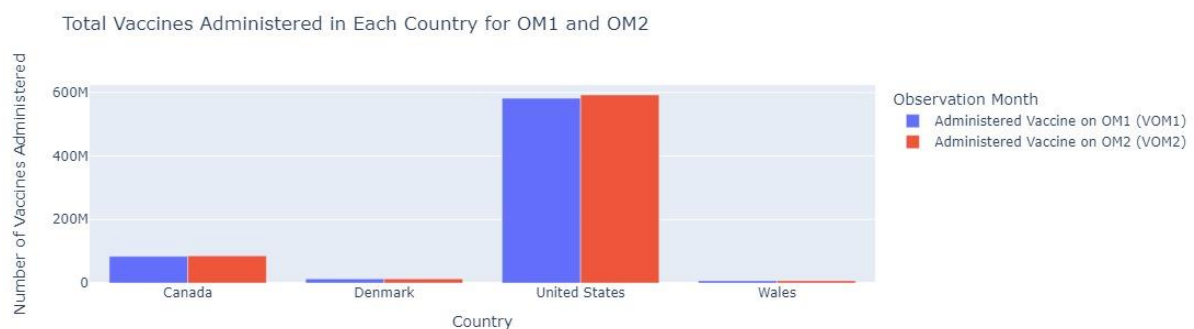
SQL Query:

```
SELECT
    '2022-04' AS "Observation Months 1 (OM1)",
    loc.location_name AS "Country Name (CN)",
    MAX(CASE
        WHEN strftime('%Y-%m', vd.date) = '2022-04'
        THEN vd.total_vaccinations ELSE 0 END)
    AS "Administered Vaccine on OM1 (VOM1)",
    '2022-05' AS "Observation Months 2 (OM2)",
    MAX(CASE
        WHEN strftime('%Y-%m', vd.date) = '2022-05'
        THEN vd.total_vaccinations ELSE 0 END)
    AS "Administered Vaccine on OM2 (VOM2)",
    MAX(CASE
        WHEN strftime('%Y-%m', vd.date) = '2022-05'
        THEN vd.total_vaccinations ELSE 0 END) -
    MAX(CASE
        WHEN strftime('%Y-%m', vd.date) = '2022-04'
        THEN vd.total_vaccinations ELSE 0 END)
    AS "Difference of totals (VOM1 - VOM2)"
FROM VaccinationData vd
JOIN Location loc ON vd.iso_code = loc.iso_code
WHERE vd.iso_code IN ('CAN', 'DNK', 'USA', 'OWID_WLS')
GROUP BY loc.location_name;
```

SQL Results:

Total rows loaded: 4						
Observation Months 1 (OM1)	Country Name (CN)	Administered Vaccine on OM1 (VOM1)	Observation Months 2 (OM2)	Administered Vaccine on OM2 (VOM2)	Difference of totals (VOM1 - VOM2)	
1 2022-04	Canada	84066328	2022-05	85472419	1406091	
2 2022-04	Denmark	12973574	2022-05	12992790	19216	
3 2022-04	United States	581969320	2022-05	592549109	10579789	
4 2022-04	Wales	7170482	2022-05	7303995	133513	

Data Visualization:



Task D.2:

```
WITH TotalMonthlyVaccinations AS (  
    SELECT  
        loc.location_name AS country_name,  
        strftime('%Y-%m', v.date) AS month_year,  
        MAX(v.total_vaccinations) AS total_vaccinations  
    FROM VaccinationData v  
    JOIN Location loc ON v.iso_code = loc.iso_code  
    WHERE v.iso_code IN ('CAN', 'DNK', 'USA', 'OWID_WLS')  
    GROUP BY loc.location_name, month_year  
,  
Average AS (  
    SELECT  
        month_year,  
        AVG(total_vaccinations) AS average_vaccinations  
    FROM TotalMonthlyVaccinations  
    GROUP BY month_year  
)  
SELECT  
    m.country_name AS "Country Name",  
    m.month_year AS "Month",  
    m.total_vaccinations AS "Cumulative Doses"  
FROM TotalMonthlyVaccinations m  
JOIN Average avg ON m.month_year = avg.month_year  
WHERE m.total_vaccinations > avg.average_vaccinations;
```

SQL Results:

Grid view		Form view	
Total rows loaded: 34			
	Country Name	Month	Cumulative Doses
1	Canada	2023-06	98238922
2	Canada	2023-07	98310269
3	Canada	2023-08	98365911
4	Canada	2023-09	98410188
5	United States	2020-12	5716312
6	United States	2021-01	38066296
7	United States	2021-02	88232689
8	United States	2021-03	172494725

Data Visualization:

Countries with More Than Average Cumulative COVID-19 Doses Administered



Task D.3:

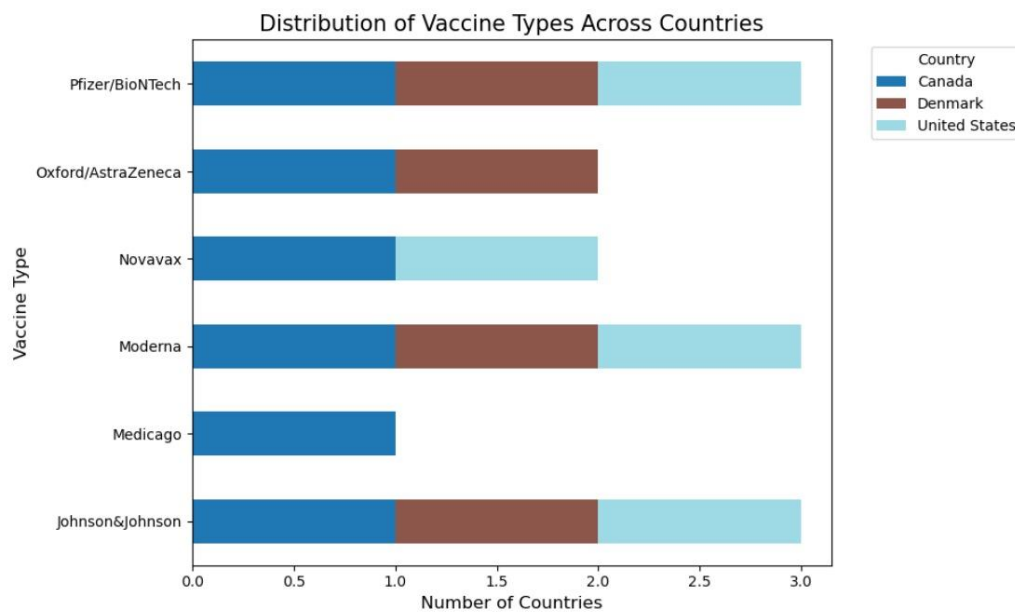
SQL Query:

```
SELECT DISTINCT
    v.vaccine_name AS "Vaccine Type",
    loc.location_name AS "Country"
FROM Vaccine v
JOIN CountryVaccinationByManufacturer cvm
    ON v.vaccine_name =
    cvm.vaccine_name
JOIN Location loc ON cvm.iso_code =
    loc.iso_code
WHERE loc.iso_code IN ('CAN', 'DNK', 'USA',
'OWID_WLS')
ORDER BY v.vaccine_name, loc.location_name;
```

SQL Results:

Grid view		Form view
		Total rows loaded: 14
	Vaccine Type	Country
1	Johnson&Johnson	Canada
2	Johnson&Johnson	Denmark
3	Johnson&Johnson	United States
4	Medicago	Canada
5	Moderna	Canada
6	Moderna	Denmark
7	Moderna	United States
8	Novavax	Canada
9	Novavax	United States

Data Visualization:



Task D.4:

SQL Query:

```
SELECT
    loc.location_name AS "Country Name",
    s.source_name AS "Source Name (URL)",
    MAX(v.total_vaccinations) AS "Total Administered Vaccines"
FROM VaccinationData v
JOIN Location loc ON v.iso_code = loc.iso_code
JOIN DataSource ds ON v.iso_code = ds.iso_code
JOIN Source s ON ds.source_id = s.source_id
WHERE loc.iso_code IN ('CAN', 'DNK', 'USA', 'OWID_WLS')
GROUP BY loc.location_name, s.source_id
ORDER BY
    MAX(v.total_vaccinations) DESC, loc.location_name,
    s.source_name;
```

SQL Results:

Grid view

Form view



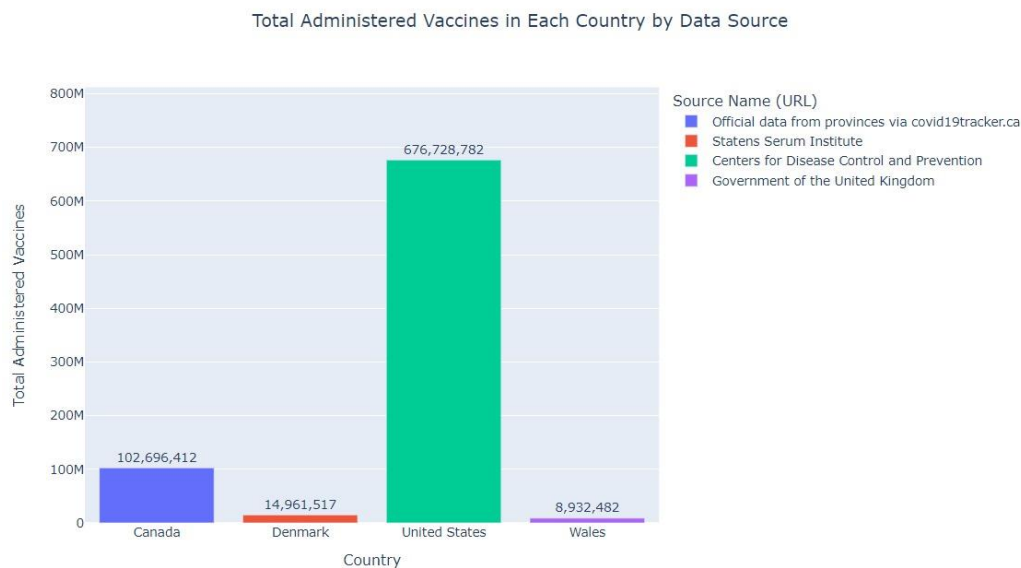
1



Total rows loaded: 4

	Country Name	Source Name (URL)	Total Administered Vaccines
1	United States	Centers for Disease Control and Prevention	676728782
2	Canada	Official data from provinces via covid19tracker.ca	102696412
3	Denmark	Statens Serum Institute	14961517
4	Wales	Government of the United Kingdom	8932482

Data Visualization:



SQL Query:

SQL Results:

Data Visualization:

